

Biped Simulation Validation Report

Topic: Self balancing and walking humanoid robot simulation which includes lower body joint and torque analysis.

By: H003,H006,H024,H029

1. Introduction

This report outlines the validation of the biped simulation created in PyBullet. The purpose is to verify that the simulated responses are consistent with Euler-Lagrange rigid-body dynamics, expected energy behaviour, and contact-based locomotion physics. The validation process uses three components: comparison with standard analytical equations, energy verification, and action-based response analysis using walking, squatting, jumping without torso twist, and jumping with torso twist.

2. Textbook Comparison

The simulation is compared against the standard Euler-Lagrange model for articulated robots.

Simulation / model form:

$$M(q)\ddot{q} + C(q, \dot{q})\dot{q} + G(q) = \tau + J_c^T \lambda$$

Textbook / analytical form:

$$\frac{d}{dt} \frac{\partial L}{\partial \dot{q}_i} - \frac{\partial L}{\partial q_i} = Q_i$$

The project implementation is consistent with the analytical structure: URDF inertial properties define the rigid-body parameters, robot.py defines the kinematic chain and Jacobian, and contact enters through ground reaction forces at the feet.

3. Energy Validation

The validation uses the standard Euler-Lagrange energy definitions.

$$T = \sum_{i=1}^n \left(\frac{1}{2} m_i v_{Ci}^2 + \frac{1}{2} \omega_i^T I_i \omega_i \right)$$

$$V = \sum_{i=1}^n m_i g z_{Ci}$$

$$E = T + V$$

The current analysis pipeline saves grouped hip, knee, and ankle kinetic, potential, and total energies for each action. These quantities provide a direct validation check on whether the simulated motion evolves in a physically meaningful way.

4. Action-Based Validation Summary

The following summaries are computed from the saved analysis CSV files in the project.

Action	Duration (s)	Peak GRF (N)	Max Hip Energy (J)	Max Knee Energy (J)	Max Ankle Energy (J)
jump w twist	9.075	2420.051669	2.888025	0.592896	0.466087
jump wo twist	9.845833	2420.051669	2.888025	0.592896	0.466087
squat	11.254167	2420.051669	2.888025	0.592896	0.466087
walk	15.108333	2420.051669	2.888025	0.592896	0.611672

Action	Hip Torque Max	Knee Torque Max	Ankle Torque Max	Hip Angle Max	Knee Angle Max	Ankle Angle Max
jump w twist	20.148115	12.727922	13.503495	1.665426	3.219316	1.611893
jump wo twist	18.985988	12.727922	14.555442	1.564331	3.245737	1.599991
squat	18.985988	12.727922	13.503495	1.793894	3.535092	1.796125
walk	18.985988	12.727922	14.590835	1.0874	1.993838	1.158896

5. Sensitivity Analysis

Sensitivity analysis for the current project is interpreted through task variation and controller/terrain parameters rather than through fluid parameters as in the reference ROV report. The biped exposes stride length, step height, and plane inclination during walking, while the action selection itself changes the required contact, torque, and energy demand.

- Walking introduces periodic contact transfer and cyclic energy exchange.
- Squatting increases grouped knee and hip excursions while keeping the feet largely planted.
- Jumping increases peak ground reaction force and transient torque demand.
- Torso twist during jumping adds additional coupled rotational demand on the lower-body chain.

6. Discussion

The validation results support the consistency of the biped simulation with Euler-Lagrange mechanics. The saved analysis confirms that the model produces finite and structured responses in ground reaction force, grouped torque, grouped angle, and grouped energy. The contact-driven nature of walking and jumping is visible through high GRF peaks, while the energy channels remain interpretable in terms of rigid-body motion and gravitational storage.

There are still modelling limitations. The current report uses a PyBullet rigid-body engine rather than a fully symbolic closed-form derivation of $M(q)$, $C(q, \dot{q})$, and $G(q)$. Contact is also numerical and depends on simulation settings rather than a dedicated analytical contact model. Even so, the system is appropriate for project-level dynamic systems analysis and validation.

7. Conclusion

The validation of the biped simulation was performed by comparing it with Euler-Lagrange dynamic structure, verifying energy behaviour, and reviewing action-dependent response trends from the saved simulation analysis

outputs. The resulting model behaves in a physically coherent manner and can be used for further educational and project-level studies in humanoid lower-body dynamics.

8. Saved Analysis Snapshot

A representative saved dashboard from the walking analysis is included below.

